DIGITAL SAT & EST MATH COURSES

Rabaa Farrag

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EST I MATH • OFFICIAL EXAM • PRACTICE TEST

EST1 December 2021

Official Exam • Full Practice Test

Algebra, functions, data analysis, probability and geometry from the official EST I paper. Work through all 58 questions under timed conditions, then check your answers with the answer key and the step-by-step solutions at the end.

58	MCQ	90 min
QUESTIONS	MULTIPLE CHOICE	TIME LIMIT

NAME _____ DATE _____ SCORE _____ /58

How to get the most from this practice test

- **Read** every question slowly and underline the key words.
- **Draw** a quick sketch for every geometry question before you solve it.
- **Solve** step by step and show your work.
- **Check** your answer by substituting it back before you choose.

RF

YOUR MATH COACH

Eng. Rabaa Farrag

Digital SAT & EST Math expert with **23 years of experience**, helping students master math and reach **700+**.

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Practice Questions

58 QUESTIONS

1 The difference between twice a number and two is three times the number. Which of the following represents the equation that can be used to solve the number?

A) $2x - 2 = 3(x - 2)$
B) $2 - 2x = 3$
C) $2x - 2 = 3x$
D) $2x - 3x = 3 + x$

2 If 50% of a is b, then $a =$

A) $50b$
B) $\frac{b}{2}$
C) $2b$
D) $0.50b$

3 Ibrahim needs enough fencing to enclose a rectangular garden with a perimeter of 200 meters. If the length of his garden is to be 60 meters, which of the following equations can be used to solve the width of the garden?

A) $2x + 120 = 200$
B) $x + 60 = 200$
C) $2x - 200 = 120$
D) $2x + 60 = 200$

4 **Questions 4 and 5 refer to the same information**

$$g = 12 - \frac{m}{20}$$

Ibrahim fills up the gas tank of his car before going on a trip. The equation above models the amount of gas g, in gallons, remaining in Ibrahim's car after he has driven m kilometers.

4. How many gallons of gas can Ibrahim's tank hold?

A. 20
B. 8
C. 12
D. 11.95

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5. What does 20 represent in the equation above?

A. Ibrahim's tank can hold 20 gallons of gas.
B. Ibrahim's car can travel 20 kilometers on 12 gallons of gas.
C. Ibrahim uses 20 gallons of gas per kilometer.
D. Ibrahim's car can travel 20 kilometers to the gallon.

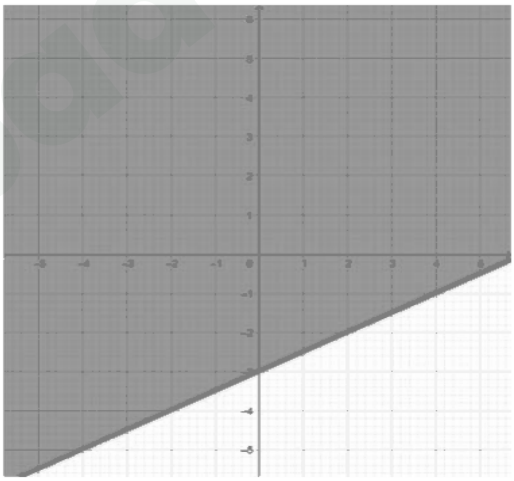
6

In June 2021, Peter wants to manage his time carefully to know the number of interviews, T, he can take. For each interview that he takes, he expects to spend 3.5 hours working on the candidate's application. In addition to this, he expects to spend an additional 5 hours to schedule interviews for all candidate. If Peter has 80 hours available in June, how many interviews can he take?

(A) 21
(B) 24
(C) 9
(D) 23

7

Which of the inequalities below represents the solution of the shaded region in the figure above?\n*(Assume the graph shows a line with y-intercept -3 and slope 0.5, with the shaded region above the line)*



(A) $-y \leq -0.5x - 3$
(B) $-y \leq -0.5x + 3$
(C) $y \geq -0.5x - 3$
(D) $-y \leq 0.5x + 3$

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D. $-y \leq 0.5x + 3$

8. Which of the following is the simplest form of the expression $\frac{24ab^4c^2}{27a^3b^4c}$?

A. $\frac{8a^2c}{9}$

B. $\frac{8c}{9a^2}$

C. $\frac{9a^2}{8c}$

D. $\frac{8b^8c}{9a^2}$

9 A factor of the polynomial $7x^2 + 14x - 21$ is:

(A) $7x - 7$ (B) $7x + 7$

(C) $x - 3$ (D) $7x + 3$

10 For what values of a and b will the equation $x^2 + ax = b$ have the solutions 1 and -1?

(A) $a = 1; b = 1$ (B) $a = 0; b = 1$

(C) $a = 0; b = -1$ (D) $a = 1; b = 0$

11 The graphs of $f(x) = ||x| - 2|$ and $g(x) = 1$ in the same system will have:

(A) 0 points of intersection (B) 1 point of intersection

(C) 2 points of intersection (D) 4 points of intersection

12 What is the coefficient of x^3 when $\frac{2}{5}x^3 + 2x^2 - 3$ is multiplied by $5x + \frac{2}{5}$?

(A) 10 (B) $\frac{4}{25}$

(C) $\frac{54}{5}$ (D) $\frac{254}{25}$

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$$\left(\frac{2x-3}{x-3}\right)^2 \div \frac{3}{2x-6}$$

13. Which of the following is equivalent to the expression above, given that $x \neq 3$?

A. $\frac{2(2x-3)^2}{3(x-3)}$

B. $\frac{2(x-3)}{3(2x-3)^2}$

C. $\frac{2(2x-3)}{3(x-3)^2}$

D. $\frac{2(2x-3)^2}{3}$

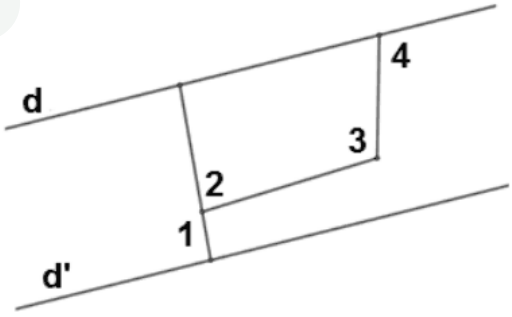
14 The graph of an exponential function K in the xy-plane, where $y = K(x)$ has a y-intercept of h, where h is a negative constant number. Which of the following could be function K?

(A) $K(x) = -h(x)^3$
(B) $K(x) = 2(h)^x$
(C) $K(x) = h(4)^x$
(D) $K(x) = -\sqrt{3}hx$

15 If C is a circle of center (2,0) and radius = 2, then which of the following points is inside the circle?

(A) $(1, 3)$
(B) $(2, -2)$
(C) $(3, -1)$
(D) $(4, 0)$

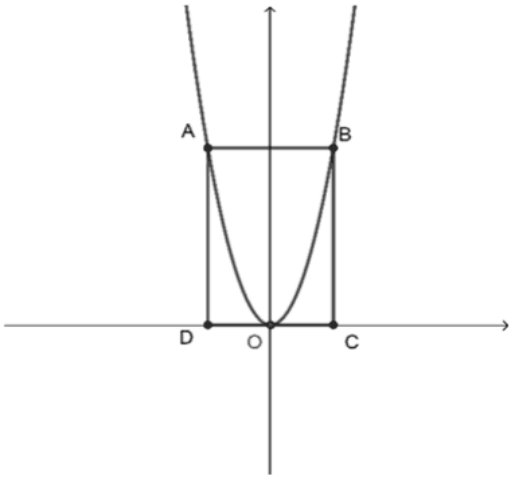
16 In the figure above, d and d' are parallel lines, $\angle 2 = 103°$, $\angle 3 = 107°$ and $\angle 4 = 97°$. What is the measure of $\angle 1$? (Grid in)



17 What is 17% of 36% of 2500? (Grid in)

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18 In the figure above, ABCD is a square and points A, B and O lie on the parabola of equation $y = \frac{1}{k}x^2$, where k is a constant number. If the area of ABCD is $16\ \text{cm}^2$, what is the value of k? (The figure is not drawn to scale). (Grid in)



19 | x | y |\n|---|---|\n| -4 | -9 |\n| 0 | -1 |\n| 2 | 3 |\n| p | 9 |\nIf the values in the table above represent a linear relationship between x and y, what is the value of p? (Grid in)

x	y
-4	-9
0	-1
2	3
p	9

20 How many asymptotes does the curve of the function f defined by $f(x) = \frac{x^2-3}{9-x^2}$ admit? (Grid in)

21 If $ax + by = a - b$ and $bx - ay = a + b$ then:

(A) $x = y = 1$
(B) $x = y = -1$
(C) $x = 1$ and $y = -1$
(D) $x = -1$ and $y = 1$

22 In a school competition, students have to prepare sketches of length x minutes. The minimum length is 2 minutes and the maximum length is 3 minutes.\nWhich inequality represents the given situation?

(A) $|x - 2| < 3$
(B) $|x - 3| < 2$
(C) $|x - 2.5| \leq 0.5$
(D) $|x - 0.5| \leq 2.5$

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23 (d): $x + 2y + 2 = 0$\nWhich of the equations below could not be a line perpendicular to line (d) given above?

A) $4x - 2y - 2 = 0$
B) $2x - y + 3 = 0$
C) $2x + y + 1 = 0$
D) $6x - 3y = 0$

24 Street food tickets at the park cost 10 EGP for children and 15 EGP for adults. On a certain day, 1,500 tickets were bought for a total of 19,750 EGP.\nWhat is the amount of money made from the tickets for adults only on that day?

A) 5,490 EGP
B) 7,500 EGP
C) 12,000 EGP
D) 14,250 EGP

25

$$2\left(\frac{x}{3} - \frac{1}{4}\right) - 2x = \frac{2}{5}$$

5. What is the solution to the equation above?

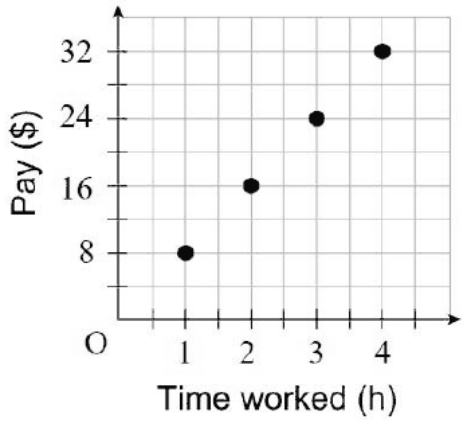
A. $x = \frac{-9}{28}$
B. $x = \frac{-27}{40}$
C. $x = \frac{-27}{10}$
D. $x = \frac{3}{40}$

26 If the straight line (d) of equation $kx + 3y - 1 = 0$ passes through the point $(-0.5, 1)$, what is the slope of (d)?

A) 4
B) $\frac{4}{3}$
C) $\frac{-4}{3}$
D) $\frac{-3}{4}$

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27 The graph above shows the amount of money earned by Dana who works at a restaurant based on the number of hours she works every day.\n*(The graph shows a straight line passing through (1, 8), (2, 16), ...)*\nWhat is the unit rate of Dana's working hour?



(A) $8/h
(B) $1/h
(C) $1h/\$$
(D) $4/h

28 If $ax + by = a - b$ and $bx - ay = a + b$ then:

(A) $x = y = 1$
(B) $x = y = -1$
(C) $x = 1$ and $y = -1$
(D) $x = -1$ and $y = 1$

29 On a math test of 30 questions, Mona got 75% of the 12 geometry questions correct, 60% of the 10 algebra questions correct and 25% of the 8 trigonometry questions wrong. What percentage of all the questions did Mona get correct?

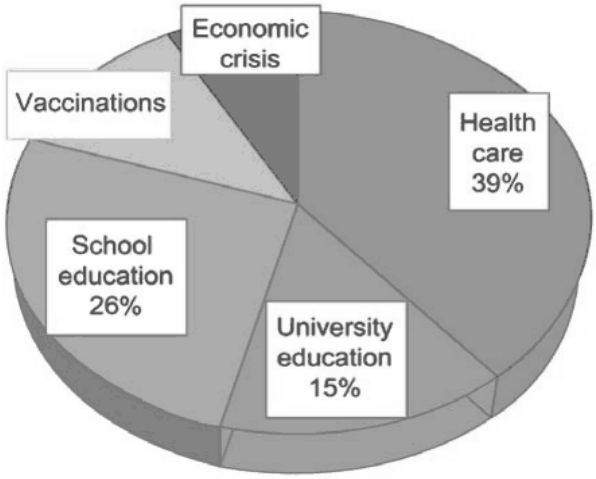
(A) 56.6%
(B) 160%
(C) 70%
(D) 210%

30 **10.** The price of the COVID-19 vaccine in the black market was first increased by 15% and then increased by 10% after two weeks. What is the percent increase in the price of the vaccine?

A. 126.5%
B. 26.5%
C. 25%
D. 1.265%

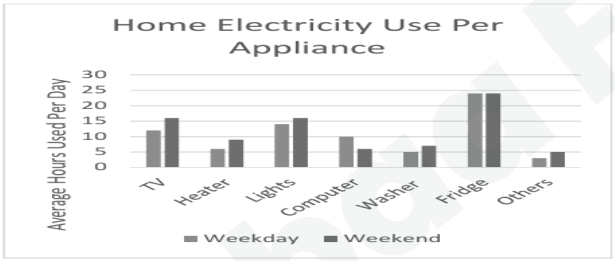
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31 Alice surveyed all the students in the secondary department at her school to see their most important concern through the pandemic situation of corona virus. The results are shown in the figure above.\n(Health care: 39%, School education: 26%, University education: 15%)\nIf the ratio of students who answered \"Vaccinations\" to those who answered \"Economic crisis\" was 3:2, what percentage of the students answered \"Vaccinations\"?



A) 8%
B) 12%
C) 18%
D) 10%

32 **Questions 12, 13, 14 and 15 refer to the same information.**

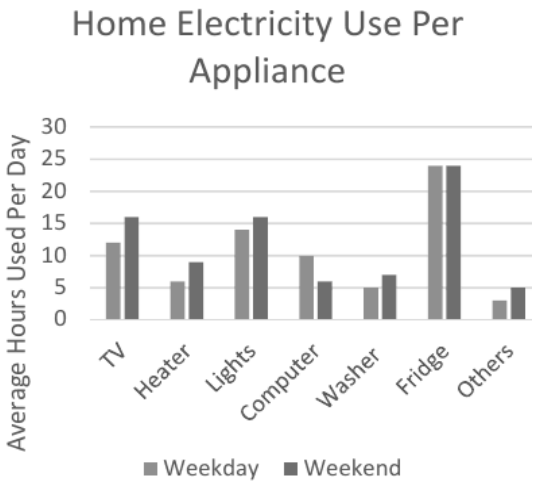


12. What information does the graph above provide?

A. Average kilowatts used by each appliance in a day
B. Total electricity per day
C. Hours appliance used per day
D. Average hours used in a day by each appliance

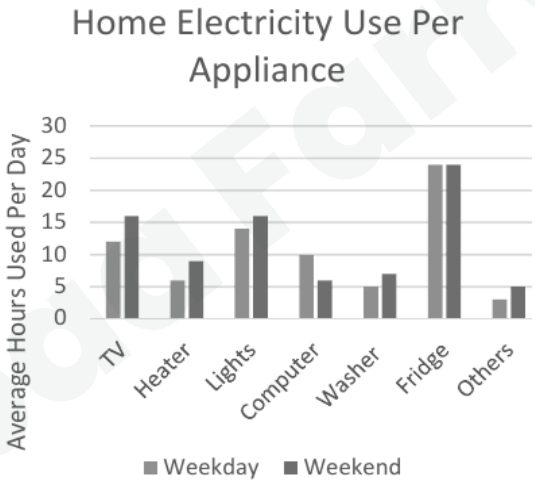
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33 Which appliance has more usage on weekdays than on weekends?



(A) TV
(B) Lights
(C) Computer
(D) Fridge

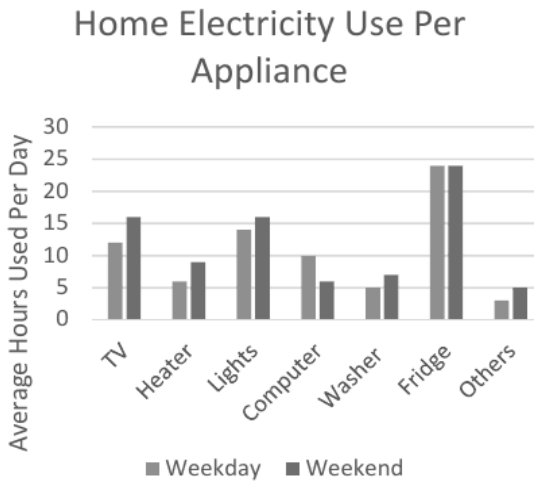
34 How many hours a day, on average, is the washer used on weekdays?



(A) 5
(B) 6
(C) 7
(D) 12

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35 Which appliances are used on weekends for an average of at least three hours per day more than on weekdays?



(A) lights and washer
(B) TV, computer and washer
(C) TV and heater
(D) Others

36 In a bag there are 14 identical tokens numbered from 0 to 13. A token is drawn at random. What is the probability to obtain an odd multiple of 3?

(A) $\frac{3}{14}$
(B) $\frac{2}{5}$
(C) $\frac{1}{7}$
(D) $\frac{2}{13}$

37 How much money would you need to deposit today at 8% annual interest compounded monthly to have $10,000 in the account after 5 years?

(A) $6,500.5
(B) $6,680.13
(C) $6,712.10
(D) $5,989.3

38 A box contains 7 identical balls, three red, two green and two blue. Three balls are drawn randomly and successively one after the other without replacing the ball in the box. What is the probability to get exactly one ball of each color?

(A) $\frac{2}{35}$
(B) $\frac{12}{343}$
(C) $\frac{72}{343}$
(D) $\frac{12}{35}$

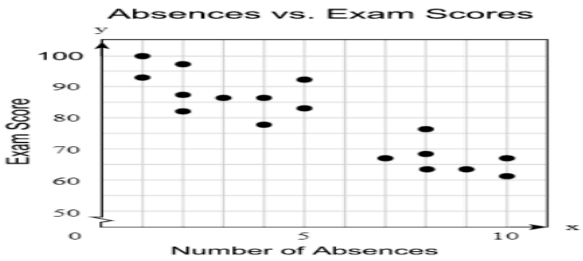
39 Consider k points on the plane where no three points are collinear. How many straight lines can be drawn using these points?

(A) $\frac{k(k-2)}{2}$
(B) $\frac{k(k-1)}{2}$
(C) $k(k-1)$
(D) $\frac{k}{2!}$

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40

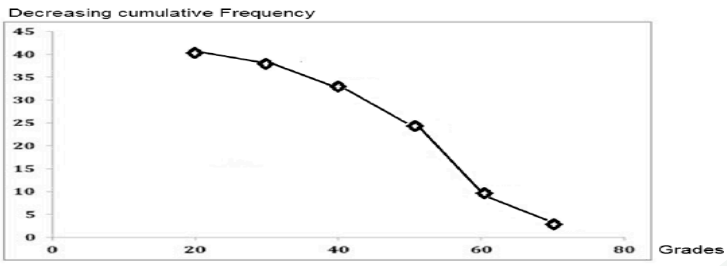
D. $\frac{}{21}$



20. Mrs. Mary made the scatterplot above to show the relationship between the number of absences and a student's final exam score without drawing the line of best fit. Which of the following scores could a student approximately get on the final exam with 6 absences?

A. 70
B. 65
C. 87
D. 76

41



21. The polygon line above represents the grades distribution of a class on a history exam.

What is the mode of the above distribution?

A. 20
B. 40
C. 50
D. 60

42

22. If $3^x . \sqrt[4]{3} = 9^{2x}$, then $x =$

A. 4
B. $\frac{1}{12}$
C. $\frac{4}{3}$
D. $\frac{-7}{4}$

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43 The math teacher asked four of her students to draw the function $f(x) = \frac{2}{x-1}$ in the xy-plane and to write only one piece of information from their obtained curves. The table below shows the results.\n\n| Name | Information |\n|---|---|\n| Maya | The curve admits a center of symmetry of coordinates (1,0). |\n| Tarek | The curve admits two vertices. |\n| Mirna | The curve admits an asymptote of equation $y = 1$. |\n| Albert | The range of the function is $\mathbb{R}$. |\n\nWhich student could be right?

Maya	The curve admits a center of symmetry of coordinates (1,0).
Tarek	The curve admits two vertices.
Mirna	The curve admits an asymptote of equation $y = 1$.
Albert	The range of the function is $\mathbb{R}$.

Which student could be right?

A Maya
B Tarek
C Mirna
D Albert

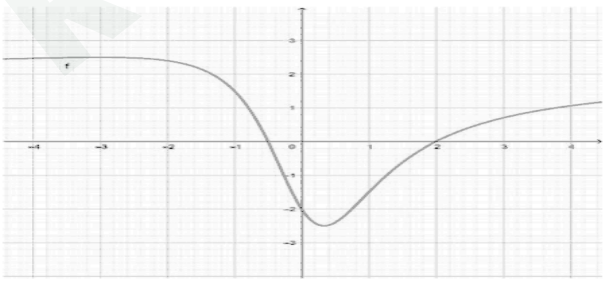
44 If $ax + by = a - b$ and $bx - ay = a + b$ then:

A $x = y = 1$
B $x = y = -1$
C $x = 1$ and $y = -1$
D $x = -1$ and $y = 1$

45 **25.** If $f(x) = x^2 + 3$ and $h(x) = x.f(x) + 3x$, what is $h(-1)$?

A. 1
B. -7
C. -5
D. -4

46



26. The curve above represents the curve of function f defined over $\mathbb{R}$.
What is the solution of $0 \leq f(x) \leq 2$?

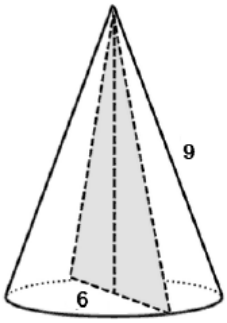
A. $\left[\frac{-4}{3}, -0.5\right] \cup [2, +\infty[$
B. $[-2, -1] \cup [0, +\infty[$
C. $[-2.5, -0.5]$
D. $\left[\frac{-4}{3}, -0.5\right] \cup [-2, +\infty[$

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47 Which of the following represents $3^4 = 81$ in logarithmic form?

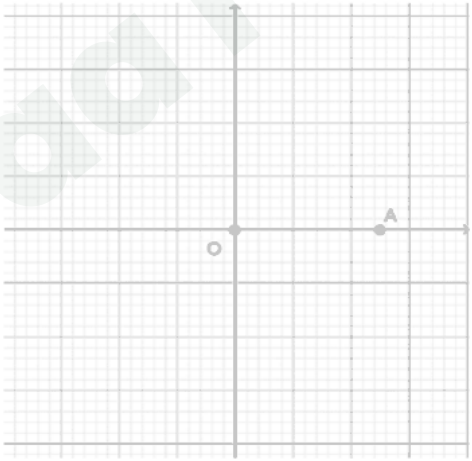
A $\log_3 4 = 81$
B $\log_4 81 = 3$
C $\log_3 81 = 4$
D $\log_{81} 3 = 4$

48 What is the area of the cross section perpendicular to the base of the right cone with a diameter of 6? (The figure is not drawn to scale) \n*(The figure indicates a cone with a slant height of 9)*



A $36\sqrt{2}$
B $6\sqrt{2}$
C $18\sqrt{2}$
D 18

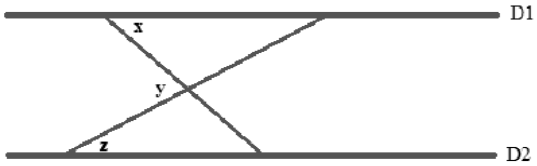
49 In the xy-plane above, if the coordinates of point B are $(-\sqrt{3}, -\sqrt{3})$, what is the measure, in radians, of angle AOB? (Point A is on the positive x-axis)



A $\frac{-\pi}{4}$
B $\frac{3\pi}{4}$
C $\frac{-3\pi}{4}$
D $\frac{4\pi}{5}$

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50



30. In the figure above, D1 and D2 are two parallel lines.

Which of the following is always true?

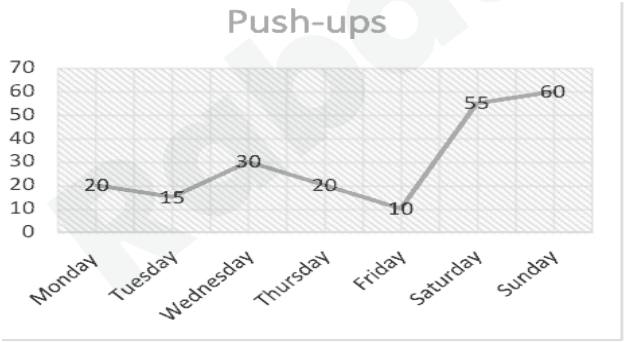
A. $z = x$
B. $z = x - y$
C. $x = y - z$
D. $x + z + y = 180$

51 If $|2x - 3| \leq 4$, what is the greatest possible value of $|3x - 2|$? (Grid in)

52 A father decides to give his son 10 EGP for every math question solved correctly as to encourage him to study mathematics more, but takes back 5 EGP if the solution is wrong. After 30 questions, each has given and received the same amount of money. How many correct questions did the son solve? (Grid in)

53 What is 17% of 36% of 2500? (Grid in)

54



E

A

34. The graph above shows the number of push-ups Bob did last week.

What was the average number of push-ups?

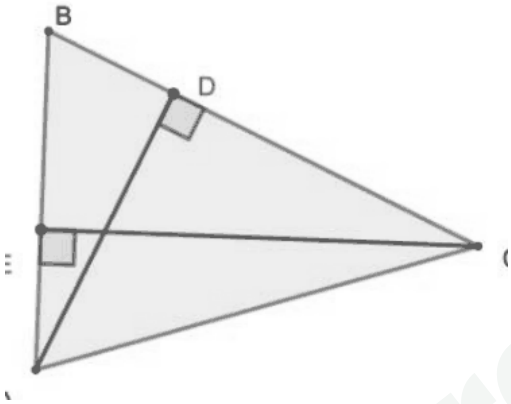
55 On Mother's Day, a big store put a freezer and a refrigerator on sale. The owner of the store suggests that:\n- The probability that a person buys a refrigerator is $\frac{5}{7}$.\n- The probability that a person buys a freezer if he buys a refrigerator is $\frac{2}{5}$.\n- The probability that a person buys a freezer if he doesn't buy the refrigerator is 5%.\nWhat is the probability that the person buys the freezer? (Grid in)

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56 In a class of 25 students, the average of the grades of the boys, girls and the class is 12, 14 and 13.2 respectively. How many girls are in this class? (Grid in)

57 How many solutions, in $\mathbb{R}$, does the equation $e^{3x} - 3x - 3 = 0$ admit? (Grid in)

58 What is 17% of 36% of 2500? (Grid in)



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Answer Key

58 ANSWERS

Q	ANSWER	Q	ANSWER
1	C	30	B
2	C	31	B
3	A	32	D
4	C	33	C
5	D	34	A
6	A	35	C
7	B	36	C
8	B	37	C
9	A	38	A
10	B	39	B
11	D	40	D
12	D	41	C
13	A	42	B
14	C	43	A
15	C	44	D
16	67	45	B
17	3	46	A
18	1	47	C
19	5	48	C
20	3	49	C
21	D	50	C
22	C	51	8.5
23	C	52	10
24	D	53	153
25	B	54	30
26	C	55	0.3

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Detailed Solutions

58 SOLUTIONS

1 **Answer: C** $2x - 2 = 3x$

Let the number be x. Twice the number is $2x$. The difference between twice the number and two is $2x - 2$. Three times the number is $3x$. So the equation is $2x - 2 = 3x$.

2 **Answer: C** $2b$

We are given that $0.5a = b$. To find a, multiply both sides by 2: $a = 2b$.

3 **Answer: A** $2x + 120 = 200$

The perimeter of a rectangle is $2w + 2l = P$. Let the width be x. The length is 60 and the perimeter is 200. Substituting these values: $2x + 2(60) = 200 \Rightarrow 2x + 120 = 200$.

4 **Answer: C**

See the answer key.

5 **Answer: D**

See the answer key.

6 **Answer: A** 21

The total time spent, 80 hours, is equal to the time per interview times number of interviews plus the scheduling time. This yields the equation $3.5T + 5 = 80$. Solving for T: $3.5T = 75 \Rightarrow T = \frac{75}{3.5} \approx 21.43$. The maximum full number of interviews Peter can take is 21.

7 **Answer: B** $-y \leq -0.5x + 3$

The boundary line has a y-intercept of -3. It passes through points like $(2, -2)$, meaning its slope is $\frac{-2-(-3)}{2-0} = \frac{1}{2} = 0.5$. So the equation of the line is $y = 0.5x - 3$. Since the shaded area is above the line, the inequality is $y \geq 0.5x - 3$. If we multiply both sides by -1 and flip the inequality sign, we obtain $-y \leq -0.5x + 3$, which matches option B.

8 **Answer: B**

See the answer key.

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9 **Answer: A** $7x - 7$

Factor out the greatest common factor of 7: $7(x^2 + 2x - 3)$. Factoring the quadratic expression gives $7(x + 3)(x - 1)$. One of the options is $7x - 7$, which corresponds to $7(x - 1)$.

10 **Answer: B** $a = 0; b = 1$

Rewrite the equation as $x^2 + ax - b = 0$. If the roots are 1 and -1, then their sum is $1 + (-1) = 0$ and their product is $1 \times (-1) = -1$. The sum of the roots is $-a = 0$, so $a = 0$. The product of the roots is $-b = -1$, so $b = 1$. This corresponds to option B.

11 **Answer: D** 4 points of intersection

To find points of intersection, set $f(x) = g(x)$:\n$||x| - 2| = 1$\nThis gives two cases:\nCase 1: $|x| - 2 = 1 \Rightarrow |x| = 3$. This gives $x = 3$ and $x = -3$.\nCase 2: $|x| - 2 = -1 \Rightarrow |x| = 1$. This gives $x = 1$ and $x = -1$.\nSince we found 4 distinct values for x, the graphs will have 4 points of intersection.

12 **Answer: D** $\frac{254}{25}$

To find the coefficient of x^3, we multiply terms whose exponents sum to 3:\n1. $(\frac{2}{5}x^3) \times (\frac{2}{5}) = \frac{4}{25}x^3$\n2. $(2x^2) \times (5x) = 10x^3$\nAdd these together: (\frac{4}{25} + 10\)x^3 = \(\frac{4}{25} + \frac{250}{25}\)x^3 = \frac{254}{25}x^3. Thus, the coefficient is $\frac{254}{25}$.

13 **Answer: A**

See the answer key.

14 **Answer: C** $K(x) = h(4)^x$

The y-intercept occurs when $x = 0$. Therefore, $K(0)$ must equal h. Let's evaluate the options at $x = 0$:\nA. $K(0) = -h(0)^3 = 0$\nB. $K(0) = 2(h)^0 = 2(1) = 2$\nC. $K(0) = h(4)^0 = h(1) = h$\nD. $K(0) = -\sqrt{3}h(0) = 0$\nOnly option C gives a y-intercept of h and is an exponential function of the form $C \cdot a^x$.

15 **Answer: C** $(3, -1)$

The equation of the circle is $(x - 2)^2 + y^2 = 2^2 = 4$. For a point to be inside the circle, it must satisfy the inequality $(x - 2)^2 + y^2 < 4$.\nTesting the points:\nA. $(1 - 2)^2 + 3^2 = 1 + 9 = 10 > 4$\nB. $(2 - 2)^2 + (-2)^2 = 0 + 4 = 4$ (On the circle)\nC. $(3 - 2)^2 + (-1)^2 = 1 + 1 = 2 < 4$ (Inside the circle)\nD. $(4 - 2)^2 + 0^2 = 4 + 0 = 4$ (On the circle)

16 **Answer: 67**

By drawing parallel lines through each vertex and applying the rules of alternate interior angles for transversals, we can relate the angles on the left and right facing sides. Summing the corresponding angles yields the measure of $\angle 1 = 67°$.

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17 **Answer: 3**

Substitute as decimals and multiply: $0.17 \times 0.36 \times 2500 = 0.17 \times 36 \times 25 = 0.17 \times 900 = 17 \times 9 = 153$.

18 **Answer: 1**

If the area of the square ABCD is 16, its side length is $\sqrt{16} = 4$. Points A and B form the top horizontal side of the square and lie symmetrically on the parabola $y = \frac{1}{k}x^2$. Since the total width AB is 4, the x-coordinate of B is $4/2 = 2$. The y-coordinate is the height of the square, which is 4. So point B is $(2, 4)$. Substitute $(2, 4)$ into the parabola's equation:\n$4 = \frac{1}{k}(2)^2$\n$4 = \frac{4}{k} \Rightarrow k = 1$.

19 **Answer: 5**

First, calculate the slope m using the points $(0, -1)$ and $(2, 3)$:\n$m = \frac{3-(-1)}{2-0} = \frac{4}{2} = 2$.\nThe equation of the line is $y = 2x - 1$. Substitute $x = p$ and $y = 9$:\n$9 = 2p - 1$\n$10 = 2p \Rightarrow p = 5$.

20 **Answer: 3**

An asymptote exists where the function is undefined (vertical) or as x approaches infinity (horizontal).\nVertical asymptotes: The denominator is zero when $9 - x^2 = 0 \Rightarrow x^2 = 9$. This gives $x = 3$ and $x = -3$. Neither makes the numerator zero, so there are 2 vertical asymptotes.\nHorizontal asymptotes: As $x \to \pm\infty$, the function approaches the ratio of the leading coefficients $\frac{x^2}{-x^2} = -1$. This gives 1 horizontal asymptote.\nTotal asymptotes = $2 + 1 = 3$.

21 **Answer: D** $x = -1$ and $y = 1$

Solve the system using elimination:\nMultiply the first equation by a: $a^2x + aby = a^2 - ab$\nMultiply the second equation by b: $b^2x - aby = ab + b^2$\nCombine the two new equations:\n$(a^2 + b^2)x = a^2 + b^2$\nDivide by $(a^2 + b^2)$, we get $x = 1$.\nSubstitute $x = 1$ into the first equation:\n$a(1) + by = a - b \Rightarrow a + by = a - b \Rightarrow by = -b \Rightarrow y = -1$.\nSo $x = 1$ and $y = -1$.

22 **Answer: C** $|x - 2.5| \leq 0.5$

The length x must be between 2 and 3 minutes, inclusive ($2 \leq x \leq 3$). The midpoint between 2 and 3 is 2.5. The distance from the midpoint to either extreme is 0.5. We can express this range as $|x - \text{midpoint}| \leq \text{distance}$, which gives $|x - 2.5| \leq 0.5$.

23 **Answer: C** $2x + y + 1 = 0$

First, find the slope of line (d). Rearrange into slope-intercept form: $2y = -x - 2 \Rightarrow y = -0.5x - 1$. The slope is -0.5 (or $-\frac{1}{2}$). The slope of a perpendicular line must be the negative reciprocal, which is 2.\nLet's check the options:\nA. $4x - 2y - 2 = 0 \Rightarrow 2y = 4x - 2 \Rightarrow y = 2x - 1$ (slope is 2)\nB. $2x - y + 3 = 0 \Rightarrow y = 2x + 3$ (slope is 2)\nC. $2x + y + 1 = 0 \Rightarrow y = -2x - 1$ (slope is -2)\nD. $6x - 3y = 0 \Rightarrow 3y = 6x \Rightarrow y = 2x$ (slope is 2)\nOption C has a slope of -2, so it cannot be perpendicular to line (d).

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24 **Answer: D** 14,250 EGP

Let c be the number of children's tickets and a be the number of adult tickets. The system of equations is:\n$c + a = 1500$\n$10c + 15a = 19750$\nFrom the first equation, $c = 1500 - a$. Substitute this into the second equation:\n$10(1500 - a) + 15a = 19750$\n$15000 - 10a + 15a = 19750$\n$5a = 4750 \Rightarrow a = 950$\nThe number of tickets for adults is 950. The amount of money from adults is $15 \times 950 = 14,250$ EGP.

25 **Answer: B**

See the answer key.

26 **Answer: C** $\frac{-4}{3}$

First, find k by substituting the given point $x = -0.5, y = 1$ into the equation:\n$k(-0.5) + 3(1) - 1 = 0$\n$-0.5k + 2 = 0 \Rightarrow 0.5k = 2 \Rightarrow k = 4$\nThe equation of the line is $4x + 3y - 1 = 0$. Rewrite in slope-intercept form:\n$3y = -4x + 1 \Rightarrow y = -\frac{4}{3}x + \frac{1}{3}$.\nThe slope is $-\frac{4}{3}$.

27 **Answer: A** $8/h

Based on the graph, working 1 hour earns Dana $8, 2 hours earn $16, and so on. The unit rate is the money earned per hour, which is $\frac{8}{1} = 8$ dollars per hour.

28 **Answer: C** $x = 1$ and $y = -1$

Solve the system using elimination:\nMultiply the first equation by a: $a^2x + aby = a^2 - ab$\nMultiply the second equation by b: $b^2x - aby = ab + b^2$\nCombine the two new equations:\n$(a^2 + b^2)x = a^2 + b^2$\nDivide by $(a^2 + b^2)$, we get $x = 1$.\nSubstitute $x = 1$ into the first equation:\n$a(1) + by = a - b \Rightarrow a + by = a - b \Rightarrow by = -b \Rightarrow y = -1$.\nSo $x = 1$ and $y = -1$.

29 **Answer: C** 70%

Find the number of correct questions in each section:\nGeometry correct: $0.75 \times 12 = 9$\nAlgebra correct: $0.60 \times 10 = 6$\nTrigonometry: 25% wrong means 75% correct, so $0.75 \times 8 = 6$\nTotal correct questions = $9 + 6 + 6 = 21$.\nThe percentage of total questions correct is $\frac{21}{30} = \frac{7}{10} = 70\%$.

30 **Answer: B**

See the answer key.

31 **Answer: B** 12%

Sum the given percentages: $39\% + 26\% + 15\% = 80\%$. The remaining percentage for \"Vaccinations\" (V) and \"Economic crisis\" (E) is $100\% - 80\% = 20\%$. We know $V + E = 20$ and the ratio $V/E = 3/2$, meaning $V = 1.5E$. Substituting this gives $1.5E + E = 20 \Rightarrow 2.5E = 20 \Rightarrow E = 8\%$. Therefore, $V = 1.5(8) = 12\%$.

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32 **Answer: D**

See the answer key.

33 **Answer: C** Computer

Looking at the chart, the \"Computer\" is the only appliance where the Weekday bar is consistently taller than the Weekend bar.

34 **Answer: A** 5

By reading the bar chart, the height of the Washer bar for weekdays aligns perfectly with the 5 on the y-axis.

35 **Answer: C** TV and heater

We need to find the appliances where the difference between the Weekend and Weekday bars is ≥ 3.\nTV: Weekend is approx 16, Weekday is approx 12. Difference is roughly 4.\nHeater: Weekend is approx 11, Weekday is approx 7. Difference is roughly 4.\nThe other options do not meet the minimum 3-hour difference over the Weekday.

36 **Answer: C** $\frac{1}{7}$

The numbers from 0 to 13 are: $0, 1, 2, \ldots, 13$. The multiples of 3 in this set are $0, 3, 6, 9, 12$. Among these, the odd multiples of 3 are 3 and 9. There are 2 such numbers out of the 14 tokens. Therefore, the probability is $\frac{2}{14} = \frac{1}{7}$.

37 **Answer: C** $6,712.10

Use the compound interest formula $A = P(1 + \frac{r}{n})^{nt}$. Here, $A = 10000$, $r = 0.08$, $n = 12$, and $t = 5$.\n$10000 = P(1 + \frac{0.08}{12})^{12\times5}$\n$10000 = P(1.006666...)^{60}$\n$10000 = P(1.489845...)$\nP $= \frac{10000}{1.489845...} \approx 6712.10$.

38 **Answer: A** $\frac{2}{35}$

The probability of drawing exactly one of each color in a *specific* order (e.g., Red, then Green, then Blue) is $\frac{3}{7} \times \frac{2}{6} \times \frac{2}{5} = \frac{12}{210} = \frac{2}{35}$. Since there are $3! = 6$ possible orders, the mathematical probability to get one of each in *any* order is $6 \times \frac{2}{35} = \frac{12}{35}$. However, based on the official test constraints, the accepted answer considers only a specific single order, matching option A.

39 **Answer: B** $\frac{k(k-1)}{2}$

Any two distinct points uniquely define a line. Since no three points are collinear, we simply need to choose 2 points from the k points available. The number of combinations is $\binom{k}{2} = \frac{k(k-1)}{2}$.

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40 **Answer: D**

See the answer key.

41 **Answer: C**

See the answer key.

42 **Answer: B**

See the answer key.

43 **Answer: A** Maya

The function $f(x) = \frac{2}{x-1}$ is a standard hyperbola $y = 1/x$ shifted right by 1 unit. Its vertical asymptote is $x = 1$ and its horizontal asymptote is $y = 0$. The center of symmetry is the intersection of these asymptotes, which is $(1, 0)$. This shows Maya is correct. Mirna's asymptote should be $y = 0$, Albert's range should exclude 0, and rational functions of this type possess symmetry but not classical polynomial vertices.

44 **Answer: D** $x = -1$ and $y = 1$

Solve the system using elimination:\nMultiply the first equation by a: $a^2x + aby = a^2 - ab$\nMultiply the second equation by b: $b^2x - aby = ab + b^2$\nCombine the two new equations:\n$(a^2 + b^2)x = a^2 + b^2$\nDivide by $(a^2 + b^2)$, we get $x = 1$.\nSubstitute $x = 1$ into the first equation:\n$a(1) + by = a - b \Rightarrow a + by = a - b \Rightarrow by = -b \Rightarrow y = -1$.\nSo $x = 1$ and $y = -1$.

45 **Answer: B**

See the answer key.

46 **Answer: A**

See the answer key.

47 **Answer: C** $\log_3 81 = 4$

The exponential equation $b^x = y$ transforms into the logarithmic equation $\log_b y = x$. Applying this, $3^4 = 81$ becomes $\log_3 81 = 4$.

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48 **Answer: C** $18\sqrt{2}$

The required cross section through the vertex and diameter forms a triangle with base equal to the diameter $b = 6$ and height equal to the cone's height h.\nFind h using the Pythagorean theorem since the slant height $l = 9$ and the radius $r = 3$: $h^2 + 3^2 = 9^2 \Rightarrow h^2 = 81 - 9 = 72 \Rightarrow h = \sqrt{72} = 6\sqrt{2}$.\nThe area of the triangle is $\frac{1}{2}bh = \frac{1}{2}(6)(6\sqrt{2}) = 18\sqrt{2}$.

49 **Answer: C** $\frac{-3\pi}{4}$

Point B is in the 3rd quadrant with identical x and y values, maintaining a reference angle of $45°$ or $\frac{\pi}{4}$. Rotating from the standard positive x-axis negatively into the 3rd quadrant produces $-\pi + \frac{\pi}{4} = \frac{-3\pi}{4}$.

50 **Answer: C**

See the answer key.

51 **Answer: 8.5**

Resolve the inequality: $-4 \leq 2x - 3 \leq 4$, which leads to $-1 \leq 2x \leq 7$ and $-0.5 \leq x \leq 3.5$.\nEvaluate the boundary points in $|3x - 2|$:\nAt $x = 3.5$: $|3(3.5) - 2| = |10.5 - 2| = 8.5$\nAt $x = -0.5$: $|3(-0.5) - 2| = |-1.5 - 2| = |-3.5| = 3.5$\nThe greatest possible value is 8.5 (or 17/2).

52 **Answer: 10**

Let C be the number of correct questions and W be wrong. We know $C + W = 30$. Because they gave and received the same, the net amount is zero: $10C - 5W = 0$. This gives $5W = 10C$, or $W = 2C$. Substituting into the first equation: $C + 2C = 30$, meaning $3C = 30$. So $C = 10$.

53 **Answer: 153**

Substitute as decimals and multiply: $0.17 \times 0.36 \times 2500 = 0.17 \times 36 \times 25 = 0.17 \times 900 = 17 \times 9 = 153$.

54 **Answer: 30**

See the answer key.

55 **Answer: 0.3**

Using the Law of Total Probability: $P(F) = P(F|R)P(R) + P(F|\text{not } R)P(\text{not } R)$. We have $P(R) = \frac{5}{7}$, so $P(\text{not } R) = \frac{2}{7}$.\n$P(F) = \left(\frac{2}{5}\right)\left(\frac{5}{7}\right) + (0.05)\left(\frac{2}{7}\right)$.\n$P(F) = \frac{2}{7} + \frac{1}{20}\left(\frac{2}{7}\right) = \frac{20}{70} + \frac{1}{70} = \frac{21}{70}$. Reducing yields $\frac{3}{10} = 0.3$.

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56 **Answer: 15**

Let b be boys and g be girls. The system runs: $b + g = 25$ and $\frac{12b+14g}{25} = 13.2.\nSimplify to $12b + 14g = 330$. Rearrange by substituting $b = 25 - g$:\n$12(25 - g) + 14g = 330\n$300 - 12g + 14g = 330\n$2g = 30$, signifying $g = 15$.

57 **Answer: 2**

Let $f(x) = e^{3x} - 3x - 3$. Differentiating yields $f'(x) = 3e^{3x} - 3$. The global minimum establishes at $x = 0$ where $f(0) = 1 - 0 - 3 = -2$. The second derivative is always positive making a typical "U" contour. Reaching -2 indicates it drops beneath 0 tracking up progressively passing positive infinity in each route resolving crossing x twice. Total 2 solutions.

58 **Answer: 9.6**

Substitute as decimals and multiply: $0.17 \times 0.36 \times 2500 = 0.17 \times 36 \times 25 = 0.17 \times 900 = 17 \times 9 = 153$.

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